AI Model Benchmark Report

A comprehensive evaluation of two Large Language Models using custom Streamlit benchmarking dashboard

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Project Overview

This benchmarking project evaluates the performance of two Large Language Models using a custom Streamlit dashboard. The system sends identical prompts to multiple models via the Hugging Face Router API and compares responses based on inference speed, output quality, and usability metrics.

The goal is to understand which model performs better for different real-world use cases including chatbots, documentation generation, and AI assistants.

Selected Models

Llama-3-8B

Provider: Meta

Parameters: 8 Billion

Type: Instruct

Link:https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct

Llama-3-70B

Provider: Meta

Parameters: 70 Billion

Type: Instruct

Link:https://huggingface.co/meta-llama/Meta-Llama-3-70B-Instruct

Llama-3-8B: The Lightweight Option

Smaller Model Size

8 billion parameters enable faster processing and lower memory footprint

Faster Inference

Reduced computational requirements for real-time applications

Lower Compute

Ideal for resource-constrained environments and edge deployment

Lightweight Applications

Suitable for chatbots, mobile AI, and customer support automation

Llama-3-70B: The Powerhouse

Larger Parameters

70 billion parameters enable deeper understanding and reasoning

Higher Accuracy

More accurate and detailed responses with better contextual awareness

Advanced Reasoning

Superior performance on complex analytical tasks and research

Enterprise Ready

Requires higher computational resources but delivers premium results

Benchmarking Methodology

Streamlit AI Benchmark Dashboard

Each model received identical prompts and results were recorded automatically using the Hugging Face Router API.

01

Inference Speed

Time taken to generate response

02

Output Quality

Word count and completeness

03

Usability

API integration ease

04

Domain Performance

Structured, meaningful answers

Model Comparison

Metric	Llama-3-8B	Llama-3-70B
Model Size	8B parameters	70B parameters
Inference Speed	Fast	Slower
Output Detail	Moderate	High
Accuracy	Good	Very High
Resource Usage	Low	High
Usability	Easy to deploy	Requires more compute

Result Summary

Speed Advantage

Llama-3-8B performed faster due to smaller parameter size, making it ideal for real-time applications requiring quick responses.

Quality Advantage

Llama-3-70B generated longer, more detailed responses with stronger reasoning and contextual awareness capabilities.

Complex Tasks

For complex prompts requiring explanation or analysis, Llama-3-70B produced significantly richer, more comprehensive responses.

Use Case Recommendations

Llama-3-8B Best For

- Chatbots and conversational AI
- Real-time assistants
- Customer support automation
- Mobile or edge AI systems
- Low-cost deployments

Llama-3-70B Best For

- Research tasks and analysis
- Technical documentation generation
- Advanced reasoning scenarios
- AI copilots for professionals
- Enterprise AI systems

Future Improvements & Conclusion

Suggestions for Enhanced Benchmarking

- **Add More Metrics**

  Response accuracy scoring, hallucination detection, token usage, cost per request

- **Compare More Models**

  Mistral 7B, Gemma 7B, Mixtral 8x7B, Claude 3 for broader understanding

- **Improve Evaluation**

  Automated datasets, multiple prompts, human evaluation, domain-specific testing

Key Takeaway: Model size strongly affects performance and speed. The choice depends on specific application requirements, balancing speed, cost, and response quality.